

KB005 - Production Deployment Issue and Rollback RCA Playbook

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1. Overview

Production deployments introduce new application code, configuration changes, infrastructure updates, and database modifications.

Deployment-related incidents occur when a recently released change negatively impacts system behavior.

Common examples:

- Application code defects
- Incorrect configuration values
- Database migration failures
- Dependency version mismatch
- Missing environment variables
- Infrastructure changes

Deployment issues commonly affect:

- Payment Gateway
- Authentication Service
- API Gateway

- Customer Services
 - Order Processing
-

2. Common Incident Symptoms

Customer Symptoms

Users may experience:

- Feature unavailable after release
 - Transactions failing
 - Login failures
 - Increased application errors
 - Slow application performance
-

System Symptoms

Common errors:

```
HTTP 500 Internal Server Error
```

```
HTTP 503 Service Unavailable
```

```
HTTP 504 Gateway Timeout
```

Application logs:

```
ERROR ApplicationStartup
```

```
Missing required configuration value:
```

```
PAYMENT_PROVIDER_ENDPOINT
```

Example:

```
NullPointerException
```

```
Application error after release v5.8.2
```

3. Deployment Investigation Metrics

Monitor:

```
application.error.rate
```

```
api.failure.percentage
```

```
deployment.timestamp
rollback.status
service.health
cpu.usage
memory.usage
```

Important comparison:

Before deployment:

Error Rate: 0.2%

Average Latency: 300ms

After deployment:

Error Rate: 25%

Average Latency: 5000ms

4. Common Root Causes and Resolution Steps

RCA Pattern 1: Application Code Defect

Description

A new release may introduce unexpected defects due to:

- Missing validations
 - Incorrect business logic
 - Runtime exceptions
 - Memory leaks
-

Symptoms

Errors start immediately after deployment.

Application logs:

```
RuntimeException
Payment calculation failed
introduced in version 5.8.2
```

Monitoring:

error_rate increased after deployment

Validation Steps

Step 1:

Check deployment timeline:

Incident Start Time:

10:05 AM

Deployment Completed:

10:00 AM

Strong correlation indicates deployment issue.

Step 2:

Review application logs:

```
grep ERROR application.log
```

Step 3:

Compare versions:

Previous:

```
payment-service:v5.7.9
```

Current:

```
payment-service:v5.8.2
```

Immediate Resolution

Actions:

1. Stop ongoing rollout.
2. Rollback application.
3. Restore previous stable version.

Example:

```
rollback payment-service --version=v5.7.9
```

1. Validate service recovery.

Permanent Fix

Engineering team should:

- Fix defect
 - Add automated tests
 - Improve release validation
-

RCA Pattern 2: Incorrect Configuration Deployment

Description

Applications depend on configuration values such as:

- Database endpoints
- API URLs
- Feature flags
- Timeout values
- Authentication settings

Wrong configuration can break production without code changes.

Symptoms

Logs:

```
ConfigurationLoadException  
Required property missing
```

or:

```
Connection refused:  
Invalid downstream endpoint
```

Validation

Compare configuration versions:

Current:

```
payment :
```

```
timeout: 100ms
```

Expected:

```
payment:
```

```
timeout: 5000ms
```

Resolution

Immediate:

- Restore previous configuration
- Restart affected services

Permanent:

- Add configuration validation
 - Implement approval workflow
-

RCA Pattern 3: Database Migration Failure

Description

Database changes during deployment may introduce failures.

Examples:

- Missing columns
 - Incorrect schema migration
 - Data corruption
 - Slow migration queries
-

Symptoms

Application errors:

```
DatabaseSchemaException
```

```
Column transaction_status not found
```

Database:

```
query failures increased after deployment
```

Validation

Check migration history:

```
SELECT *  
FROM migration_history  
ORDER BY executed_time DESC;
```

Verify schema:

```
DESCRIBE transactions;
```

Immediate Resolution

Actions:

1. Stop application rollout.
 2. Restore compatible schema.
 3. Rollback application version.
-

Permanent Fix

Implement:

- Backward compatible migrations
 - Automated migration testing
 - Rollback scripts
-

RCA Pattern 4: Dependency Version Compatibility Issue

Description

New deployments may upgrade:

- Libraries
- APIs
- External dependencies

Version mismatch can create failures.

Symptoms

Logs:

```
MethodNotFoundException
```

```
Unsupported API version
```

or:

```
External service contract mismatch
```

Validation

Compare:

Current dependency:

```
payment-sdk:v3.0
```

Expected:

```
payment-sdk:v2.5
```

Resolution

Immediate:

- Rollback dependency version
- Restore previous application build

Permanent:

- Add compatibility testing
 - Maintain dependency contracts
-

RCA Pattern 5: Feature Flag Issue

Description

Feature flags enable controlled releases.

Incorrect activation can expose incomplete features.

Symptoms

Only some users affected:

Examples:

```
10% customers seeing checkout failure
```

```
New UI users affected only
```

Validation

Check active feature flags:

```
{  
  "new_payment_flow": true  
}
```

Review rollout percentage:

```
enabled_percentage = 10%
```

Resolution

Immediate:

Disable feature:

```
{  
  "new_payment_flow": false  
}
```

Permanent:

- Improve feature validation
 - Add monitoring per feature
-

5. Deployment Incident Investigation Checklist

Step 1: Confirm Deployment Correlation

Questions:

- When did issue start?
 - Was there a release recently?
 - Which version changed?
-

Step 2: Check Deployment Changes

Review:

- Code commits
 - Configuration updates
 - Database migrations
 - Dependency updates
-

Step 3: Decide Rollback

Rollback recommended when:

- Customer impact is high
- Root cause unknown
- Error rate continues increasing

Rollback not required when:

- Small configuration fix available
 - No customer impact
-

Step 4: Validate Recovery

After rollback confirm:

```
error_rate reduced
latency normal
transactions successful
health checks passing
```

6. Escalation Matrix

Issue Type	Escalation Team
Application Bug	Development Team
Configuration Failure	DevOps Team
Database Migration Issue	Database Engineering
Infrastructure Change	Cloud Operations
Release Pipeline Issue	Release Engineering

7. Related Historical Incidents

INC-50120

Issue:

Payment Gateway failures started after deployment.

Root Cause:

Incorrect timeout configuration released.

Resolution:

Configuration rollback completed.

INC-50442

Issue:

Users unable to login after authentication release.

Root Cause:

JWT library version incompatibility.

Resolution:

Rolled back authentication service.

INC-50991

Issue:

Order API returning HTTP 500 errors.

Root Cause:

Database schema migration issue.

Resolution:

Restored previous schema version.

8. RCA Generation Template

Deployment RCA reports should include:

Root Cause

Identify the exact change responsible.

Evidence

Include:

- Deployment timestamp
- Error timeline
- Logs
- Configuration comparison

Resolution Applied

Recovery action performed.

Preventive Actions

Long-term release improvements.

Example:

Root Cause:

Payment Gateway outage occurred because incorrect timeout configuration was deployed.

Evidence:

- Incident started 5 minutes after deployment.
- Error rate increased from 0.2% to 25%.
- Similar incident INC-50120 identified.

Resolution:

Rolled back configuration to previous stable version.

Preventive Action:

Added automated configuration validation before release.

End of Document